The Master Equation of Threshold Dispatch: A Theory of Waiting in Capacity-Constrained Service

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Abstract

Every capacity-constrained service operation — a matatu at a stage, a parcel van on a route, an intercity coach, a container ship, a cloud compute pool — must decide when to depart. Too early, and the departure is unprofitable because the fixed cost is not amortised over enough units. Too late, and the customer abandons because the wait exceeds their tolerance. This paper derives the **master equation** of threshold dispatch, which states the maximum waiting time as a function of three primitives: the cost structure, the arrival rate, and the service time. The equation decomposes waiting into a demand term and a state term, with the cost structure acting as a multiplicative factor. From this single equation we derive: (i) the vehicle-size irrelevance proposition, which shows that in the absence of endogenous congestion the vehicle size does not determine the amount of waiting; (ii) the regime switch, which determines whether a corridor is demand-constrained or state-constrained; (iii) the demand adequacy condition, a hard floor on arrival rate below which no corridor is viable; and (iv) the waiting conservation principle, which states that the time is fixed by the parameters and the design choice is only where it is held. We then extend the model to endogenous congestion, tolerance-constrained dispatch, heterogeneous units, parallel buckets, and subsidy necessity, and derive the two-gate diagnostic that the practitioner should follow. The paper is purely theoretical. Its contribution is to isolate the one equation from which the operational implications of the consolidation theory follow, and to state precisely when it holds and when it fails.

1 Introduction

Consider the operation of a vehicle that can carry up to K units of demand and costs c_0 to dispatch, regardless of how full it is. Each unit earns a net margin of $p - c_v$, where p is the price and c_v is the variable cost. If the vehicle dispatches with N units, its contribution is

$$\Pi(N) = N\,(p - c_v) - c_0. \tag{1}$$

The vehicle is profitable if and only if $N \geq N^*$, where

$$N^* = \frac{c_0}{p - c_v} \tag{2}$$

is the threshold quantity. Below N^*, the departure loses money; above it, the departure earns money. The threshold load factor $\lambda^* = N^*/K$ is the fraction of capacity that must be filled for the departure to break even.

This is exactly the **consolidation invariant** of Papers 0 and 3 of this series, restated for a single departure. What those papers did not isolate — and what this paper adds — is the *temporal* consequence of the invariant. A departure must wait for N^* units to accumulate before it can fire. That waiting is not a side effect of cost; it is a computed quantity, and it is the quantity every customer experiences.

Now add a customer. The customer cares about one thing: how long they wait between the moment they arrive and the moment they are served. If the vehicle waits to fill to N^*, the customer waits. If the vehicle departs early, it loses money. The tension between the customer's patience and the vehicle's viability is the subject of this paper.

The central question is this: **given a fixed cost, a margin per unit, an arrival rate, and a service time per unit, how long must the customer wait?** The answer is a single equation, and its structure has implications that reach further than the operational case that motivated it.

We proceed as follows. Section 2 sets up the model. Section 3 derives the master equation. Section 4 establishes the structural properties: the decomposition, the vehicle-size irrelevance, the regime switch, the demand adequacy condition, the waiting conservation principle, and the optimality of threshold dispatch. Section 5 extends the model to endogenous service time (congestion), tolerance-constrained dispatch, heterogeneous units, parallel buckets, and subsidy necessity. Section 6 states the two-gate diagnostic that follows. Section 7 describes the limits of the model.

2 Setup

2.1 The departure

Definition 1 (Departure)**.** *A **departure** is a single dispatch of a vehicle that carries $N \in [0, K]$ units and incurs a fixed cost $c_0 > 0$. Each unit served earns a net margin $m = p - c_v > 0$. Contribution is*

$$\Pi(N) = Nm - c_0. \tag{3}$$

Definition 2 (Threshold quantity)**.** *The **threshold quantity** is*

$$N^* = \frac{c_0}{m}, \tag{4}$$

the smallest N at which $\Pi(N) \geq 0$.

Definition 3 (Threshold load factor)**.** *The **threshold load factor** is*

$$\lambda^* = \frac{N^*}{K} = \frac{c_0}{Km}, \tag{5}$$

the fraction of capacity that must be filled for the departure to break even.

Remark 1 (The threshold quantity is independent of capacity)**.** *N^* is independent of K. It depends only on the cost structure (c_0, m). The threshold load factor λ^*, by contrast, depends on K inversely: larger vehicles require a lower load factor to reach the same threshold quantity. This separation — the quantity is fixed by cost, the load factor by cost over capacity — is the hinge of the vehicle-size irrelevance result below.*

2.2 The bucket

Definition 4 (Bucket)**.** *A **bucket** is the set of units that will be carried on a single departure. The bucket accumulates at rate μ, the arrival rate of demand in the corridor.*

This is the object Paper 5 showed to be a bulk-service queue: a maximum batch size K, a minimum batch size $\lceil \lambda^* K \rceil$, and a fixed service (startup) cost c_0. The kill rule of that paper is the "N-policy" of queueing theory. The present paper changes the question: instead of asking when the departure is *profitable*, it asks when the departure becomes *possible in time*.

2.3 Arrivals and service time

Assumption 1 (Deterministic arrivals)**.** *Demand arrives at a constant rate μ units per unit time. We relax this to stochastic arrivals in Section 7.*

Assumption 2 (Independent service time)**.** *Each unit takes a service time τ with mean $\bar{\tau} > 0$. Service times are independent across units. We take $\bar{\tau}$ as exogenous in Sections 2–4 and endogenise it in Section 5.*

The mean service time $\bar{\tau}$ is the reciprocal of the **service rate** R of Paper 0 and Paper 1: $\bar{\tau} = 1/R(a)$, where a is locatability. This paper deliberately writes $\bar{\tau}$ rather than $R(a)$ to keep the full generality of the result free of the addressability instantiation; Section 5 and the companion strategy essay reinstate the three-way decompostion of $\bar{\tau}$ into addressability, infrastructure, and congestion.

2.4 The customer

Definition 5 (Customer tolerance)**.** T_{tol} *is the maximum waiting time a customer will accept before abandoning the queue.*

Definition 6 (Maximum waiting time)**.** $T_{\max}$ *is the length of the delivery window the firm must promise, measured from the arrival of the first unit in the bucket to the delivery of the last unit.*

The firm is viable for a given customer if and only if

$$T_{\max} \leq T_{\text{tol}}. \tag{6}$$

2.5 The dispatch rule

Assumption 3 (Threshold dispatch)**.** *The firm dispatches at $N = N^*$: it waits exactly until the bucket reaches the threshold quantity, then departs.*

This is the myopic rule that maximises loading efficiency subject to viability. In Section 4 we show it is optimal when the tolerance constraint is slack. In Section 5 we consider the tolerance-constrained case where the firm may be forced to depart at $N < N^*$.

3 Derivation of the Master Equation

Fix a dispatch at $N = N^*$. The bucket accumulates at rate μ, so the time to fill is

$$T_{\text{fill}} = \frac{N^*}{\mu} = \frac{c_0}{m\,\mu}. \tag{7}$$

After the bucket fills, the vehicle serves N^* units sequentially at a rate of $1/\bar{\tau}$ per unit, so the time to serve is

$$T_{\text{serve}} = N^*\,\bar{\tau} = \frac{c_0\,\bar{\tau}}{m}. \tag{8}$$

The maximum waiting time — from the arrival of the first unit to the delivery of the last — is the sum:

$$T_{\max} = T_{\text{fill}} + T_{\text{serve}} = \frac{c_0}{m\,\mu} + \frac{c_0\,\bar{\tau}}{m}. \tag{9}$$

Theorem 1 (Master equation)**.** *Under Assumptions 1–2 and the threshold dispatch rule, the maximum waiting time satisfies*

$$\boxed{T_{\max} = \frac{c_0}{p - c_v}\left[\frac{1}{\mu} + \bar{\tau}\right]} \tag{10}$$

where c_0 is the fixed cost per departure, $p - c_v$ is the margin per unit, μ is the arrival rate, and $\bar{\tau}$ is the mean service time per unit.

Proof. Direct from the derivation above, with $m = p - c_v$. □

Definition 7 (Waiting multiplier)**.** *The* ***waiting multiplier*** *is*

$$\Gamma = \frac{c_0}{p - c_v}. \tag{11}$$

The master equation can then be written

$$T_{\max} = \Gamma \left[\frac{1}{\mu} + \bar{\tau} \right]. \tag{12}$$

Remark 2 (The multiplier carries the market)**.** *The margin $p - c_v$, and with it the multiplier Γ, is not itself a routing quantity. It is the market's determination — the price at which possession changes hands, less the cost of the unit. The master equation therefore couples the two halves of the routing layer at a single point: the market supplies p, and the equation tells the mover how long, at that price and that cost structure, a customer must wait before a departure is possible.*

Remark 3 (The multiplier is the threshold quantity)**.** *The waiting multiplier Γ is nothing other than the threshold quantity N^* of Definition 2. The master equation therefore says something precise and new:* the number of units you must accumulate to break even is exactly the number that scales your customers' waiting. *The consolidation invariant of Papers 0 and 3 was a profitability condition; the master equation is its temporal shadow, and the two share a single primitive, Γ.*

4 Structural Properties

4.1 The decomposition

Proposition 1 (Demand–state decomposition)**.** *The maximum waiting time decomposes additively into a demand term and a state term, with a common multiplicative factor:*

$$T_{\max} = \Gamma \cdot T_{\text{demand}} + \Gamma \cdot T_{\text{state}}, \tag{13}$$

where

$$T_{\text{demand}} = \frac{1}{\mu}, \qquad T_{\text{state}} = \bar{\tau}. \tag{14}$$

The demand term is the headway (the inverse of the arrival rate). The state term is the mean service time, inflated by addressability, infrastructure, and congestion.

Proof. Immediate from Theorem 1. □

Corollary 1 (Demand as the base of the waiting)**.** *The demand term is a lower bound on the total waiting:*

$$T_{\max} \geq \frac{\Gamma}{\mu}. \tag{15}$$

Equality holds as $\bar{\tau} \to 0$. No amount of state improvement (reduction of $\bar{\tau}$) can bring $T_{\max}$ below Γ/μ.

Proof. Since $\bar{\tau} > 0$, $T_{\max} = \Gamma(1/\mu + \bar{\tau}) > \Gamma/\mu$. □

This is the formal statement of the claim that **demand sits at the base of waiting**. The demand term Γ/μ is the floor; the state term $\Gamma\bar{\tau}$ builds on it. If demand is thin (small μ), the floor is high, and no state improvement can reduce the total below it. This is the formal anchor for the strategic rule that client acquisition precedes technology build.

4.2 Vehicle-size irrelevance

Proposition 2 (Vehicle-size irrelevance)**.** *The maximum waiting time is independent of the vehicle capacity K.*

Proof. K appears in the threshold load factor $\lambda^* = N^*/K$ but cancels from the threshold quantity $N^* = c_0/m$. The master equation depends on N^* only, and therefore on K not at all. □

Corollary 2 (Waiting is set by cost, not capacity)**.** *For a fixed corridor (fixed μ) and a fixed state (fixed $\bar{\tau}$), replacing a small vehicle with a larger one does not change the waiting time. It changes the* load factor *at which the vehicle operates, but not the* quantity *that must be accumulated before dispatch, and therefore not the* time*.*

Remark 4 (The conditional nature of the result)**.** *Proposition 2 holds only when $\bar{\tau}$ is exogenous. When service time depends on congestion, which depends on the number of vehicles on the road, the result fails — but through the tolerance-constrained regime, not the threshold regime. This is the subject of Section 5.1, where the failure is derived precisely.*

This proposition is the complement, not the contradiction, of the capacity-choice analysis of Paper 3. Paper 3 asks *which K to buy* given a corridor's density; the present proposition says that, *once the cost structure (c_0, m) is fixed,* the amount a customer must wait to make a departure viable does not itself depend on K. The two results jointly imply the strategy: choose K for the corridor (Paper 3), and price the waiting for the cost structure (this paper) — they are separate decisions.

4.3 The regime switch

Definition 8 (Demand–service ratio)**.** *The* ***demand–service ratio*** *of a corridor is*

$$\rho = \mu\,\bar{\tau} = \frac{\mu}{1/\bar{\tau}} = \frac{\textit{arrival rate}}{\textit{service rate}}. \tag{16}$$

Proposition 3 (Regime switch)**.** *The relative weight of the demand and state terms in the master equation is given by ρ:*

$$\frac{T_{\text{demand}}}{T_{\text{state}}} = \frac{1/\mu}{\bar{\tau}} = \frac{1}{\rho}. \tag{17}$$

Therefore:

1. *If $\rho < 1$, the demand term dominates. The corridor is* ***demand-constrained****.*
2. *If $\rho = 1$, the terms are balanced.*
3. *If $\rho > 1$, the state term dominates. The corridor is* ***state-constrained****.*

Proof. Immediate from the definitions and the master equation. □

Remark 5 (The queueing interpretation)**.** *The ratio $\rho = \mu\bar{\tau}$ is the traffic intensity of the underlying queue. When $\rho < 1$, arrivals are slower than service, the queue remains empty on average, and the bucket fills slowly — the vehicle is waiting for demand. When $\rho > 1$, arrivals are faster than service, the queue grows, and the service time dominates — the demand is waiting for the vehicle. The regime switch is the* stability boundary *of the underlying service system, reinterpreted as a diagnostic for which constraint binds. This is the generalisation of the utilisation diagnostic of Paper 0 ($u = R_{\text{obs}}/R(a)$): that ratio measured loss against an addressability ceiling; ρ measures demand against service, and is the first thing to check.*

4.4 The demand adequacy condition

Proposition 4 (Demand adequacy)**.** *A corridor is viable if and only if the arrival rate satisfies*

$$\mu \geq \mu_{\min} = \frac{c_0}{T_{\text{tol}}(p - c_v) - c_0\,\bar{\tau}} = \frac{\Gamma}{T_{\text{tol}} - \Gamma\bar{\tau}}. \tag{18}$$

Proof. From $T_{\max} \leq T_{\text{tol}}$:

$$\frac{c_0}{p - c_v}\left[\frac{1}{\mu} + \bar{\tau}\right] \leq T_{\text{tol}} \tag{19}$$

$$\frac{1}{\mu} + \bar{\tau} \leq \frac{T_{\text{tol}}(p - c_v)}{c_0} = \frac{T_{\text{tol}}}{\Gamma} \tag{20}$$

$$\frac{1}{\mu} \leq \frac{T_{\text{tol}} - \Gamma\bar{\tau}}{\Gamma} \tag{21}$$

$$\mu \geq \frac{\Gamma}{T_{\text{tol}} - \Gamma\bar{\tau}}. \tag{22}$$

□

Corollary 3 (Structural infeasibility)**.** *If $T_{\text{tol}} \leq \Gamma\bar{\tau}$, then no arrival rate is high enough to satisfy the demand adequacy condition. The corridor is* ***structurally infeasible****: no operator, at any price, at that cost structure, can serve it within tolerance.*

Proof. If $T_{\text{tol}} \leq \Gamma\bar{\tau}$, then the denominator $T_{\text{tol}} - \Gamma\bar{\tau} \leq 0$, and the inequality in Proposition 4 cannot be satisfied for any $\mu > 0$. □

The structural infeasibility condition has a clean interpretation: $\Gamma = c_0/m$ is the threshold quantity N^*, and $T_{\text{tol}}/\bar{\tau}$ is the maximum number of units that could be served within tolerance (if fill time were zero). The corridor is infeasible when the threshold quantity exceeds the maximum serviceable quantity:

$$N^* > \frac{T_{\text{tol}}}{\bar{\tau}}. \tag{23}$$

Corollary 4 (Comparative statics of $\mu_{\min}$)**.** *The minimum arrival rate is:*

- *increasing in c_0 ($\partial\mu_{\min}/\partial c_0 > 0$): higher fixed cost requires more demand.*
- *decreasing in p ($\partial\mu_{\min}/\partial p < 0$): higher price reduces the demand requirement.*
- *increasing in c_v ($\partial\mu_{\min}/\partial c_v > 0$): higher variable cost requires more demand.*
- *increasing in $\bar{\tau}$ ($\partial\mu_{\min}/\partial\bar{\tau} > 0$): slower service requires more demand.*
- *decreasing in T_{tol} in absolute value ($\partial\mu_{\min}/\partial T_{\text{tol}} < 0$): more tolerance reduces the requirement.*

Proof. Direct differentiation of the expression in Proposition 4. □

4.5 Waiting conservation

Proposition 5 (Waiting conservation)**.** *For fixed parameters $(c_0, m, \mu, \bar{\tau})$, the maximum waiting time $T_{\max}$ is conserved. It can be transferred between the customer, the vehicle, the driver, the capital, and the subsidy, but it cannot be eliminated by any reallocation of the cost of holding it.*

Proof. $T_{\max}$ depends only on the four parameters. Changing who holds the waiting does not change the parameters, and therefore does not change $T_{\max}$. □

Remark 6 (The waiting is not the cost)**.** *Proposition 5 states that the* time *is conserved. It does not state that the* cost *is conserved. The cost of holding the waiting depends on where it is held: a waiting parcel costs storage, a waiting passenger costs their time, a waiting driver costs their wage, a waiting vehicle costs its depreciation, and waiting capital costs the rate of interest. The design choice is not "how much waiting" but "who bears the cost of holding it." This is the central strategic claim of the companion essay (*Where Does the Waiting Go?*, Paper 26), stated here in its formal form.*

4.6 Optimality of threshold dispatch

Proposition 6 (Optimal dispatch)**.** *Under the threshold dispatch rule, the firm's profit is non-negative. Any dispatch at $N < N^*$ produces a loss. Any dispatch at $N > N^*$ delays the customer without increasing profit per unit. The threshold N^* is therefore the unique profit-maximising dispatch quantity subject to non-negativity when the tolerance constraint is slack.*

Proof. $\Pi(N) = Nm - c_0$ is increasing in N with slope $m > 0$. At $N = N^*$, $\Pi = 0$. Below N^*, $\Pi < 0$. Above N^*, $\Pi > 0$ and increasing. But dispatching above N^* requires waiting for more units, which increases $T_{\max}$ without a corresponding requirement from the customer (who only cares about $T_{\max} \leq T_{\text{tol}}$). So the firm's incentive is to dispatch at exactly N^* when μ is high enough, or at the tolerance-bound quantity $N^{\text{opt}} = \min(K, N^{\text{tol}}_{\max})$ when tolerance binds (Section 5.2). □

5 Extensions

5.1 Endogenous congestion

The vehicle-size irrelevance proposition (Proposition 2) assumed exogenous $\bar{\tau}$. When $\bar{\tau}$ depends on congestion, which depends on the number of vehicles on the road, the proposition fails — but through a specific mechanism, derived here.

Definition 9 (Traffic flow)**.** *The traffic flow is*

$$f = \frac{\mu}{N^*} = \frac{\mu\,(p - c_v)}{c_0}, \tag{24}$$

the number of departures per unit time on the corridor.

Proposition 7 (Endogenous service time)**.** *Suppose the service (trip) time depends on traffic flow through the standard BPR congestion function*

$$\bar{\tau}(f) = \bar{\tau}_0 \left[1 + \alpha \left(\frac{f}{F_c}\right)^{\beta}\right], \tag{25}$$

where $\bar{\tau}_0$ is free-flow service time, F_c is road capacity, and $\alpha, \beta > 0$. Then:

1. *For $\beta > 0$, $\bar{\tau}$ is increasing in f, and therefore increasing in μ.*
2. *The master equation becomes implicit in μ:*

$$T_{\max} = \Gamma \left[\frac{1}{\mu} + \bar{\tau}_0 \left(1 + \alpha \left(\frac{\mu}{F_c N^*}\right)^{\beta}\right)\right]. \tag{26}$$

3. *Under* threshold *dispatch, the dispatched quantity is fixed at N^*, which is independent of K. The frequency $f = \mu/N^*$ is therefore also independent of K: congestion is independent of vehicle size in the threshold regime.*

Proof. (i) is direct from the congestion function, which is increasing in f. (ii) substitutes $\bar{\tau}(f)$ with $f = \mu/N^*$ into the master equation. (iii) follows because $N^* = c_0/m$ contains no K; the vehicle-size effect cannot enter through f while the departure is pinned at the threshold quantity. □

Remark 7 (Where the vehicle-size effect does operate)**.** *The vehicle-size effect on congestion appears only when the dispatch quantity is* not *fixed at the threshold. In the* tolerance-constrained *case (Section 5.2), the firm dispatches at* $N^{\text{opt}} = \min(K, N^{\text{tol}}_{\max})$*. When* $K \leq N^{\text{tol}}_{\max}$ *the vehicle fills to capacity and frequency is* μ/K*; when* $K \geq N^{\text{tol}}_{\max}$ *the vehicle is tolerance-constrained and frequency is* $\mu/N^{\text{tol}}_{\max}$*. The larger the vehicle, the lower the frequency, and the lower the congestion. Congestion therefore discriminates against small vehicles* only *in the tolerance-constrained regime, and this is the substantive content of the congestion trap derived in the companion essay.*

5.2 Tolerance-constrained dispatch

Proposition 8 (Tolerance-constrained dispatch)**.** *When the tolerance constraint is binding, the firm dispatches at*

$$N^{\text{opt}} = \min\left(K,\ N^{\text{tol}}_{\max}\right), \tag{27}$$

where

$$N^{\text{tol}}_{\max} = \frac{T_{\text{tol}}\,\mu}{1+\rho}. \tag{28}$$

The firm is viable only if $N^{\text{opt}} \geq N^*$*, which is the demand adequacy condition.*

Proof. The constraint $T_{\max}(N) \leq T_{\text{tol}}$ gives $N(1/\mu + \bar{\tau}) \leq T_{\text{tol}}$, so $N \leq T_{\text{tol}}/(1/\mu + \bar{\tau}) = T_{\text{tol}}\,\mu/(1+\rho) = N^{\text{tol}}_{\max}$. The capacity constraint gives $N \leq K$. The firm chooses the largest feasible N, i.e., $\min(K, N^{\text{tol}}_{\max})$. Viability requires this to be at least N^*. □

Corollary 5 (Frequency and congestion)**.** *The dispatch frequency is*

$$f = \frac{\mu}{N^{\text{opt}}} = \begin{cases} \mu/K & \text{if } K \leq N^{\text{tol}}_{\max}, \\ (1+\rho)/T_{\text{tol}} & \text{if } K > N^{\text{tol}}_{\max}. \end{cases} \tag{29}$$

Large vehicles operate in the second regime, where frequency is pinned by tolerance rather than by vehicle capacity. Small vehicles operate in the first regime, where frequency is set by capacity.

This is the key to the congestion trap. Tolerance forces a maximum dispatch size. If vehicles are small, they fill to capacity and dispatch at frequency μ/K. If vehicles are large, they cannot fill to capacity without violating tolerance, so they dispatch at the tolerance-bound quantity and frequency $\mu/N^{\text{tol}}_{\max}$. Since $N^{\text{tol}}_{\max}$ may be much larger than a small vehicle's capacity, the large vehicle dispatches *less frequently*, reducing congestion. Consolidation reduces congestion by allowing vehicles to be larger than tolerance alone would justify, but not so large that they cannot fill within tolerance.

5.3 Heterogeneous units

Definition 10 (Unit classes)**.** *Suppose demand arrives in* J *classes, each with its own arrival rate* μ_j*, margin* m_j*, and service time* $\bar{\tau}_j$*. The total arrival rate is* $\mu = \sum_j \mu_j$*.*

Proposition 9 (Heterogeneous master equation)**.** *Under threshold dispatch with heterogeneous units and a single departure per class, the maximum waiting time is*

$$T_{\max} = \max_j \left\{ \Gamma_j \left[\frac{1}{\mu_j} + \bar{\tau}_j \right] \right\}, \tag{30}$$

where $\Gamma_j = c_0/m_j$ *is the class-specific waiting multiplier.*

Proof. Each class must reach its own threshold $N_j^* = c_0/m_j$ before its departure is profitable. The class-specific waiting time is $T_{\max,j} = \Gamma_j(1/\mu_j + \bar{\tau}_j)$. If the firm promises a single window to all classes, the window must accommodate the longest class waiting, hence the maximum. □

Corollary 6 (Class selection)**.** *The firm should reject classes for which $T_{\max,j} > T_{\text{tol}}$ unless the class can be consolidated with another class to reduce the effective $\bar{\tau}$ or increase the effective μ.*

This gives the formal basis for the client-selection rule of the Peleka thesis: the firm should prefer clients whose properties (high μ_j, low $\bar{\tau}_j$) produce a $T_{\max,j}$ within tolerance, and reject those whose properties produce a $T_{\max,j}$ outside it. Repeatability is preferred not for its own sake but because it is a high-μ_j, low-$\bar{\tau}_j$ class.

5.4 Parallel buckets

Definition 11 (Parallel buckets)**.** *Suppose the firm operates M independent buckets in parallel, each on a separate corridor, with its own arrival rate μ_j and service time $\bar{\tau}_j$.*

Proposition 10 (Parallel waiting)**.** *The total waiting time across all buckets is*

$$T_{\max}^{\text{total}} = \max_j T_{\max,j}, \tag{31}$$

if the firm makes a single promise across all corridors, or

$$T_{\max}^{\text{total}} = \frac{1}{M} \sum_j T_{\max,j}, \tag{32}$$

if each corridor has its own promise. The corridor-specific waiting times are unchanged from the single-bucket case.

Proof. If the firm makes one promise, it is bound by the worst corridor. If it makes corridor-specific promises, the customer experiences the waiting of their own corridor, and the average is the arithmetic mean. □

Corollary 7 (The theory is per-corridor)**.** *Parallel operation does not change the per-corridor waiting. It only changes the aggregation. The theory is entirely a per-corridor theory, and parallel operation is a portfolio of per-corridor decisions — which is precisely the architecture of the Peleka Express design (Paper 27), where each route-corridor is a bucket and the network is a portfolio of buckets.*

5.5 Subsidy necessity

Definition 12 (Scheduled departure)**.** *A **scheduled** departure is one whose departure time is fixed in advance and does not depend on the bucket quantity. The firm commits to depart at time t_0 regardless of $N(t_0)$.*

Proposition 11 (Subsidy necessity)**.** *For a scheduled departure at time t_0 with expected dispatch quantity $N(t_0) = \mu t_0$, the expected contribution is*

$$\Pi^{\text{sched}} = \mu t_0 \cdot m - c_0. \tag{33}$$

If $\mu t_0 < N^*$, the scheduled departure loses money at the rate

$$S = c_0 - \mu t_0\, m = m(\Gamma\mu - \mu t_0). \tag{34}$$

This loss must be recovered by a price premium, a merge with another departure, or a subsidy.

Proof. Direct calculation from $\Pi(N) = Nm - c_0$ at $N = \mu t_0$. □

Corollary 8 (The subsidy is the price of the schedule)**.** *When the model refuses to wait for the threshold (as scheduled departures must), the waiting is not eliminated. It is held by the public purse as a subsidy, or by the paying passengers as a price premium, or it is eliminated by merging departures. There is no fourth option.*

The subsidy per departure is

$$S = \max(0,\ c_0 - \lambda K m)\,, \tag{35}$$

where λ is the realised load factor at departure. This is the sixth place the waiting can go, after the parcel, the passenger, the driver, the vehicle, and the capital. It is the formal counterpart of the scheduled-transit inversion analysed in Paper 18: schedule protection removes the waiting from the vehicle, and the gap it leaves is a claim on the public purse.

6 The Two-Gate Diagnostic

The master equation and its extensions yield a sequential diagnostic procedure.

6.1 The first gate: demand adequacy

Proposition 12 (Demand gate)**.** *Before diagnosing which state dimension binds, the corridor must pass the demand gate:*

$$D = \frac{\mu}{\mu_{\min}} \geq 1. \tag{36}$$

If $D < 1$, the corridor cannot be served profitably at that vehicle class and price, and no state improvement will change that. The remedy is to change the vehicle class, the price, or the corridor.

Proof. Direct from Proposition 4. □

6.2 The second gate: state decomposition

If the demand gate is passed, the utilisation ratio decomposes into three state components:

$$U_a = \frac{R_{\text{observed}}}{R(\text{actual } a, \text{best } I, \text{best } C)} \tag{37}$$

$$U_I = \frac{R(\text{actual } a, \text{actual } I, \text{best } C)}{R(\text{actual } a, \text{best } I, \text{best } C)} \tag{38}$$

$$U_C = \frac{R(\text{actual } a, \text{actual } I, \text{actual } C)}{R(\text{actual } a, \text{actual } I, \text{best } C)} \tag{39}$$

where R is the service rate $1/\bar{\tau}$ at the indicated state (addressability a, infrastructure I, congestion C). The three ratios multiply to the overall utilisation $U = U_a \cdot U_I \cdot U_C$. Whichever is lowest is the binding state constraint.

Proposition 13 (Two-gate sequence)**.** *The diagnostic is a sequence, not a ratio:*

1. *Check D. If $D < 1$, the corridor is demand-inadequate. Change vehicle, price, or corridor.*
2. *If $D \geq 1$, compute the state decomposition. Whichever of U_a, U_I, U_C is lowest determines the binding state dimension and the appropriate remedy.*

Proof. The demand gate is a precondition: if $\mu < \mu_{\min}$, no state improvement brings $T_{\max}$ below tolerance. The state decomposition is only meaningful once the gate is passed. □

Corollary 9 (The wrong lever is the expensive error)**.** *Investing in a state dimension that is not binding — verifying addresses where demand is thin, buying larger vehicles where infrastructure caps capacity, adding vehicles where congestion is the constraint — cannot reduce total waiting below its binding floor. The two-gate sequence exists to prevent exactly this error, and its position in the diagnostic is worth the entire rest of the procedure.*

7 Limits of the Model

The master equation is derived under specific assumptions, and its scope is bounded by them.

Deterministic arrivals. Assumption 1 treats μ as a constant. Under stochastic arrivals, the waiting time becomes a random variable, and the relevant quantity is a quantile (e.g., the 95th percentile) rather than the maximum. The master equation then becomes a bound on the expected waiting time, and the quantile bound requires the variance of μ and τ. This is a natural extension, and the qualitative conclusions (demand as base, vehicle-size irrelevance under exogenous τ, demand adequacy condition) all survive.

Exogenous service time. Assumption 2 treats $\bar{\tau}$ as a parameter. When $\bar{\tau}$ is endogenous to congestion (Section 5.1), the master equation becomes an implicit equation in μ and requires solving for the fixed point. The vehicle-size irrelevance fails in this case, and the failure is the substantive content of the congestion trap.

Single departure per class. The threshold dispatch assumption means the firm operates one bucket at a time. Extensions to parallel buckets (Section 5.4) and to multiple classes (Section 5.3) preserve the structure of the equation but change the aggregate waiting. The per-corridor theory is fully captured by the single-bucket model; the portfolio theory requires the extensions.

No capacity choice. The model treats K as given. It does not derive the optimal K; that is the subject of Paper 3 (capacity choice) and Paper 6 (the infrastructure ceiling on K). The present model is a theory of waiting for a given vehicle, not a theory of fleet composition.

No price discrimination. The model uses a single price p for all units. In practice, the firm may offer priority classes, advance-purchase discounts, and dynamic pricing — the tier structure of the Peleka Universal Rate Model (Paper 9). Each of these changes the effective μ and m per class, and therefore the effective waiting multiplier. The extensions in Section 5.3 are a step toward price discrimination but not a full treatment.

No exogenous competition. The model assumes the firm is the only operator in the corridor. Under competition, the customer's tolerance T_{tol} is set by the competitor, not by the firm. The analysis is unchanged in structure, but T_{tol} becomes an endogenous variable in the competitive game.

Central recommendation. The master equation is a theory of *threshold dispatch in isolation*. Its implications are strongest for corridors where the firm is the only operator, the arrivals are regular, and the service time is exogenous. Its implications weaken precisely where its assumptions weaken, and the theory should be applied with those bounds in mind.

8 Conclusion

The master equation is

$$T_{\max} = \frac{c_0}{p - c_v}\left[\frac{1}{\mu} + \bar{\tau}\right]. \tag{40}$$

It has three primitives: the cost structure $c_0/(p - c_v)$, the arrival rate μ, and the service time $\bar{\tau}$. From these three, it determines the maximum waiting time a customer must accept, and therefore the maximum delivery window the firm can promise.

The equation has five implications that structure the operational decision.

First, waiting decomposes into demand and state terms. The demand term $1/\mu$ is the base; the state term $\bar{\tau}$ builds on it. This is why no state improvement can reduce waiting below the demand floor: the floor is set by demand.

Second, vehicle size is irrelevant (ceteris paribus). The threshold quantity $N^* = c_0/m$ is independent of K, and the waiting time depends only on N^*. Vehicle size affects the load factor but not the waiting. This is a conditional result, and its failure under endogenous congestion is the substance of the congestion trap.

Third, the regime switch determines which constraint binds. The ratio $\rho = \mu\bar{\tau}$ separates the demand-constrained regime from the state-constrained regime. The diagnostic is the ratio, not either term alone.

Fourth, the demand adequacy condition is a hard floor. Below $\mu_{\min}$, no corridor is viable at that vehicle class and price. This is a precondition for the diagnostic, not a lever.

Fifth, waiting is conserved. The time is fixed by the parameters; the design choice is only where it is held. The five places (parcel, passenger, driver, vehicle, capital) plus the sixth (subsidy) are the routes for a conserved quantity.

The equation is simple, and it is enough. It says that the waiting a logistics business holds is not a mystery and not a matter of judgement. It is a computed quantity, determined by the cost structure, the arrival rate, and the service time. The business that prices this quantity survives; the business that ignores it is eventually repriced by it.

> **The master equation in one sentence:** the maximum waiting is the cost structure multiplied by the sum of the headway and the service time, and every implication of the theory — vehicle-size irrelevance, regime switch, demand adequacy, waiting conservation — follows from reading that equation carefully.